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Chapter Review

Problems: Additional Problems

29. S Consider an infinite number of identical particles, each with charge q, placed along the x axis at distances a, $2a$, $3a$, $4a$, . . . from the origin. What is the electric field at the origin due to this distribution? *Suggestion:* Use

$$1+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+\cdots=\frac{\pi^2}{6}$$

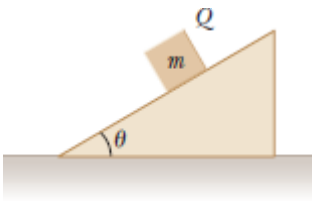
30. A particle with charge $-3.00\,\text{nC}$ is at the origin, and a particle with negative charge of magnitude Q is at $x = 50.0\ \text{cm}$. A third particle with a positive charge is in equilibrium at $x = 20.9\ \text{cm}$. What is Q?

31. AMT A small block of mass m and charge Q is placed on an insulated, frictionless, inclined plane of angle θ as in Figure P22.31. An electric field is applied parallel to the incline.

(a) Find an expression for the magnitude of the electric field that enables the block to remain at rest.

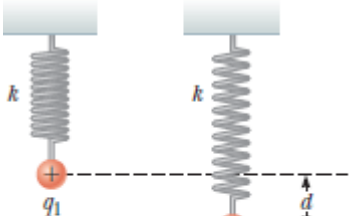
(b) If $m = 5.40\ \text{g}$, $Q = -7.00\ \mu\text{C}$, and $\theta = 25.0°$, determine the magnitude and the direction of the electric field that enables the block to remain at rest on the incline.

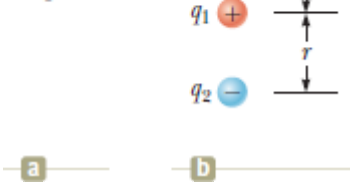
Figure P22.31



32. A small sphere of charge $q_1 = 0.800\ \mu\text{C}$ hangs from the end of a spring as in Figure P22.32a. When another small sphere of charge $q_2 = -0.600\ \mu\text{C}$ is held beneath the first sphere as in Figure P22.32b, the spring stretches by $d = 3.50\ \text{cm}$ from its original length and reaches a new equilibrium position with a separation between the charges of $r = 5.00\ \text{cm}$. What is the force constant of the spring?

Figure P22.32





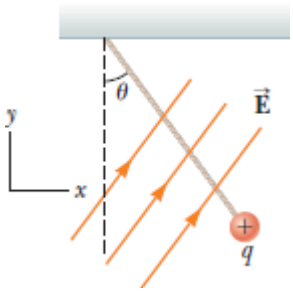
33. NS A charged cork ball of mass 1.00 g is suspended on a light string in the presence of a uniform electric field as shown in Figure P22.33. When $\vec{\mathbf{E}} = (3.00\hat{\mathbf{i}} + 5.00\hat{\mathbf{j}}) \times 10^5 \text{ N/C}$, the ball is in equilibrium at $\theta = 37.0°$. Find

(a) the charge on the ball and

(b) the tension in the string.

Figure P22.33

Problems 33 and 34.



34. NS A charged cork ball of mass m is suspended on a light string in the presence of a uniform electric field as shown in Figure P22.33. When $\vec{\mathbf{E}} = A\hat{\mathbf{i}} + B\hat{\mathbf{j}}$, where A and B are positive quantities, the ball is in equilibrium at the angle θ. Find

(a) the charge on the ball and

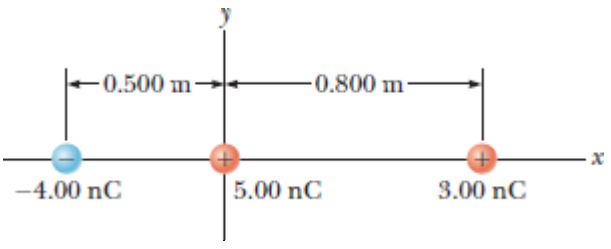
(b) the tension in the string.

35. Three charged particles are aligned along the x axis as shown in Figure P22.35. Find the electric field at

(a) the position $(2.00 \text{ m}, 0)$ and

(b) the position $(0, 2.00 \text{ m})$.

Figure P22.35



36. QC Two point charges $q_A = -12.0\ \mu\text{C}$ and $q_B = 45.0\ \mu\text{C}$ and a third particle with unknown charge q_C are located on the x axis. The particle q_A is at the origin, and q_B is at $x = 15.0 \text{ cm}$. The third particle is to be placed so that each particle is in equilibrium under the action of the electric forces exerted by the

other two particles.

(a) Is this situation possible? If so, is it possible in more than one way? Explain. Find

(b) the required location and

(c) the magnitude and the sign of the charge of the third particle.

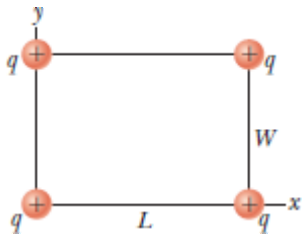
37. Two small spheres hang in equilibrium at the bottom ends of threads, 40.0 cm long, that have their top ends tied to the same fixed point. One sphere has mass 2.40 g and charge $+300\,\mathrm{nC}$. The other sphere has the same mass and charge $+200\,\mathrm{nC}$. Find the distance between the centers of the spheres.

38. Four identical charged particles $(q = +10.0\ \mu\mathrm{C})$ are located on the corners of a rectangle as shown in Figure P22.38. The dimensions of the rectangle are $L = 60.0\ \mathrm{cm}$ and $W = 15.0\ \mathrm{cm}$. Calculate

(a) the magnitude and

(b) the direction of the total electric force exerted on the charge at the lower left corner by the other three charges.

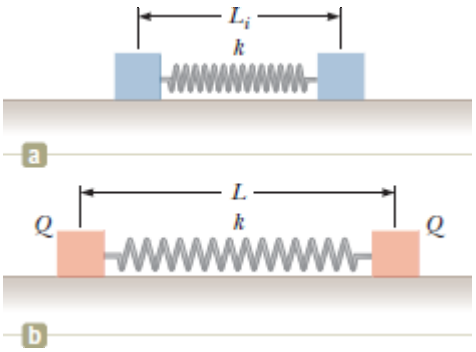
Figure P22.38



39. NS **Review**. Two identical blocks resting on a frictionless, horizontal surface are connected by a light spring having a spring constant $k = 100\ \mathrm{N/m}$ and an unstretched length $L_i = 0.400\ \mathrm{m}$ as shown in Figure P22.39a. A charge Q is slowly placed on each block, causing the spring to stretch to an equilibrium length $L = 0.500\ \mathrm{m}$ as shown in Figure P22.39b. Determine the value of Q, modeling the blocks as charged particles.

Figure P22.39

Problems 39 and 40.

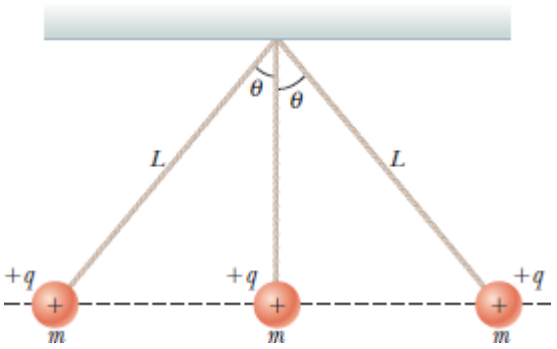


40. NS **Review**. Two identical blocks resting on a frictionless, horizontal surface are connected by a light spring having a spring constant k and an unstretched

length L_i as shown in Figure P22.39a. A charge Q is slowly placed on each block, causing the spring to stretch to an equilibrium length L as shown in Figure P22.39b. Determine the value of Q, modeling the blocks as charged particles.

41. Three identical point charges, each of mass $m = 0.100\ \text{kg}$, hang from three strings as shown in Figure P22.41. If the lengths of the left and right strings are each $L = 30.0\ \text{cm}$ and the angle θ is $45.0°$, determine the value of q.

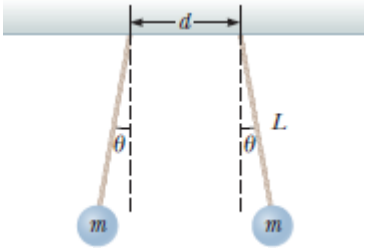
Figure P22.41



42. *Why is the following situation impossible?* An electron enters a region of uniform electric field between two parallel plates. The plates are used in a cathode-ray tube to adjust the position of an electron beam on a distant fluorescent screen. The magnitude of the electric field between the plates is 250 N/C. The plates are 0.200 m in length and are separated by 3.50 cm. The electron enters the region at a speed of $3.50 \times 10^6\ \text{m/s}$, traveling parallel to the plane of the plates in the direction of their length. It leaves the plates heading toward its correct location on the fluorescent screen.

43. Two hard rubber spheres, each of mass $m = 15.0\ \text{g}$, are rubbed with wool on a dry day and are then suspended with two insulating strings of length $L = 5.00\ \text{cm}$ whose support points are a distance $d = 3.00\ \text{cm}$ from each other as shown in Figure P22.43. During the rubbing process, one sphere receives exactly twice the charge of the other. They are observed to hang at equilibrium, each at an angle of $\theta = 10.0°$ with the vertical. Find the amount of charge on each sphere.

Figure P22.43

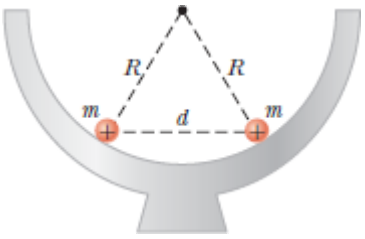


44. S Two identical beads each have a mass m and charge q. When placed in a hemispherical bowl of radius R with frictionless, nonconducting walls, the beads move, and at equilibrium, they are a distance d apart (Fig. P22.44).

(a) Determine the charge q on each bead.

(b) Determine the charge required for d to become equal to $2R$.

Figure P22.44



45. QIC S Two small spheres of mass m are suspended from strings of length ℓ that are connected at a common point. One sphere has charge Q and the other charge $2Q$. The strings make angles θ_1 and θ_2 with the vertical.

(a) Explain how θ_1 and θ_2 are related.

(b) Assume θ_1 and θ_2 are small. Show that the distance r between the spheres is approximately

$$r \approx \left(\frac{4k_e Q^2 \ell}{mg} \right)^{1/3}$$

46. CR You are working as an expert witness for an inventor. The inventor devised a system that allows an 85.0-kg human to hover above the ground at the surface of the Earth due to the repulsive force between a charge q applied to his body and the normal electric charge on the Earth. The normal charge on the Earth is such that the electric field is uniform near the Earth's surface, directed downward toward the surface, and is of magnitude 130 N/C at the location of the engineer's experiments. Everything went well until the engineer tried a new experiment. He attempted to transfer the same amount of charge q to each of *two* experimental subjects standing next to each other, so they could hover and work close together on a task. The charged, hovering experimental subjects repelled each other and were injured as they flew away in opposite directions. Both experimental subjects are now suing the inventor for their injuries. The inventor is claiming that it is not his fault if the subjects find each other repulsive. To find out whether the inventor has a good defense, determine the initial acceleration of each subject if they are working 1.00 m apart.

47. QIC **Review**. A 1.00-g cork ball with charge $2.00\ \mu\text{C}$ is suspended vertically on a 0.500-m-long light string in the presence of a uniform, downward-directed electric field of magnitude $E = 1.00 \times 10^5\ \text{N/C}$. If the ball is displaced slightly from the vertical, it oscillates like a simple pendulum.

(a) Determine the period of this oscillation.

(b) Should the effect of gravitation be included in the calculation for part (a)? Explain.